

Rational and Irrational Numbers on the Number Line

MA.8.NSO.1.2

I can order and compare any set of real numbers, and plot them.

Every number you have met this unit has one place on the line.

Today is deciding which place, and which of two is greater.

Warm-Up

On your own. Four minutes.

Same and different

What is the same about these two, and what is different?

$$\sqrt{17}$$

$$-\sqrt[3]{17}$$

Notes

I. Every real number has exactly one place on the line

- A. Rational or irrational, fraction or root or decimal — one number, one point.
- B. The form it is written in does not move it. $\sqrt{4}$ and 2 are the same point.

Notes

II. Turn every number into a decimal first

- A. Compare decimals, then write the answer back in the original form.
- B. Two places is enough for everything on this worksheet.
- C. A root that comes out even is not an estimate — write it exactly.

Notes

III. With negatives, further left is smaller

- A. -2 is greater than -5 , even though 5 is greater than 2 .
- B. So the bigger the root, the further LEFT a negative sits.
- C. $-\sqrt{50}$ is less than $-\sqrt{49}$. **The radicand went up and the value went down.**

Example 1

Nothing on paper yet. Answer out loud.

$$\sqrt{3} \quad -\frac{11}{5} \quad \pi \quad -\sqrt[3]{50}$$

1. Which of the four are irrational?
2. Which one is furthest left, before you compute anything?
3. Which two are you least sure about?

Example 1

Now the decimals. Two places each.

$$\sqrt{3}$$

$$-\frac{11}{5}$$

$$\pi$$

$$-\sqrt[3]{50}$$

Fill these in before you order anything.

Example 1

The decimals, checked.

$$\sqrt{3} \quad -\frac{11}{5} \quad \pi \quad -\sqrt[3]{50}$$

$$1.73 \quad -2.20 \quad 3.14 \quad -3.68$$

Example 1

Least to greatest.

$$-\sqrt[3]{50}$$

$$-\frac{11}{5}$$

$$\sqrt{3}$$

$$\pi$$

$$-3.68 < -2.20 < 1.73 < 3.14$$

Example 2

Marisol's work. Something is wrong.

“ $-\sqrt{50}$ is greater than -7 , because 50 is greater than 49.”

Find the error.

Example 2

Name the error.

1. $50 > 49$ — true.
2. $\sqrt{50} > \sqrt{49}$ — true.
3. $-\sqrt{50} > -7$ — false. The negative sign reverses it.

Example 2

Fix it.

$$-\sqrt{50} \quad \text{vs} \quad -7$$
$$-7.071 \quad \text{vs} \quad -7.000$$

$$-\sqrt{50} < -7$$

Whiteboards • Question 1 of 9

Boards up on three.

$$-\sqrt{16} \qquad -4$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 1 of 9

Answer.

$$-\sqrt{16}$$

$$-4$$

$$-4.000 \quad \text{vs}$$

$$-4.000$$

Answer: =

Whiteboards • Question 2 of 9

Boards up on three.

$$-\sqrt{17} \qquad -4$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 2 of 9

Answer.

$$-\sqrt{17}$$

$$-4$$

$$-4.123 \text{ vs}$$

$$-4.000$$

Answer: <

Whiteboards • Question 3 of 9

Boards up on three.

$$-\sqrt{17}$$

$$-4.1$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 3 of 9

Answer.

$$-\sqrt{17}$$

$$-4.1$$

$$-4.123 \text{ vs}$$

$$-4.100$$

Answer: <

Whiteboards • Question 4 of 9

Boards up on three.

$$\sqrt{17} \qquad 4.1$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 4 of 9

Answer.

$$\sqrt{17}$$

4.1

4.123

vs

4.1

Answer: >

Whiteboards • Question 5 of 9

Boards up on three.

$$-\sqrt[3]{17} \qquad -2.6$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 5 of 9

Answer.

$$\begin{array}{ccc} -\sqrt[3]{17} & & -2.6 \\ -2.571 & \text{vs} & -2.600 \end{array}$$

Answer: >

Whiteboards • Question 6 of 9

Boards up on three.

$$-\sqrt[3]{17} \qquad -2.5$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 6 of 9

Answer.

$$\begin{array}{ccc} -\sqrt[3]{17} & & -2.5 \\ -2.571 & \text{vs} & -2.500 \end{array}$$

Answer: <

Whiteboards • Question 7 of 9

Boards up on three.

$$-\sqrt{50} \qquad -7$$

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 7 of 9

Answer.

$$-\sqrt{50}$$

$$-7$$

$$-7.071 \text{ vs}$$

$$-7.000$$

Answer: <

Whiteboards • Question 8 of 9

Boards up on three.

π 3.14

Compare the two numbers.

Write <, > or =.

Whiteboards • Question 8 of 9

Answer.

π

3.14

3.14159...

vs

3.14000

Answer: >

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Jess says $-\sqrt{17}$ is greater than -4 , because 17 is greater than 16. Is she right?

Write your answer in sentences.

This one is written work. Say which part of her reasoning is correct.

Whiteboards • Question 9 of 9

Answer.

Jess says $-\sqrt{17}$ is greater than -4 , because 17 is greater than 16. Is she right?

$-\sqrt{17} = -4.123$, which is LEFT of -4

Answer: No — it is less

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.